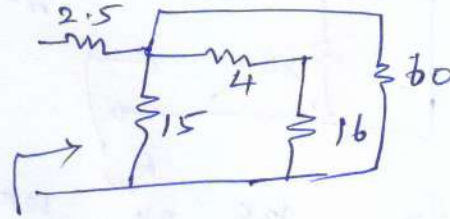
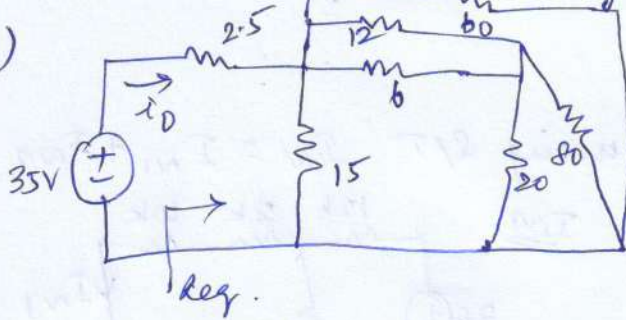


Slot: C2 key.

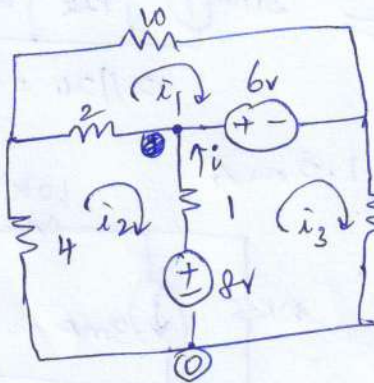
1)



$$R_{eq} = 2.5 + 15 \parallel (20 + 60)$$

$$R_{eq} = 10 \Omega$$

2)



loop 1: $6 = 12i_1 - 2i_2 \Rightarrow 6i_1 - i_2 = 3$ — (1)

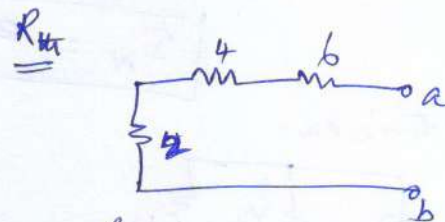
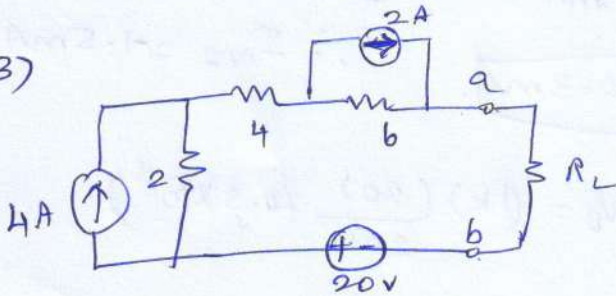
loop 2: $-8 = -2i_1 + 7i_2 - i_3$ — (2)

loop 3: $-8 + 6 + 6i_3 - i_2 = 0 \Rightarrow -i_2 + 6i_3 = 2$ — (3)

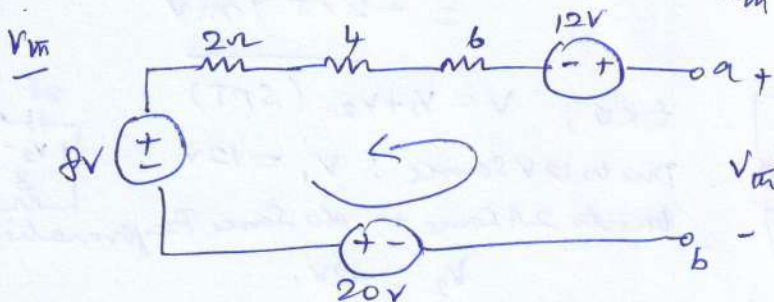
Solving for $i_1, i_2 + i_3$

at node (0) $i_1 + i_2 = i_3$

3)



$$R_{th} = 12 \Omega$$

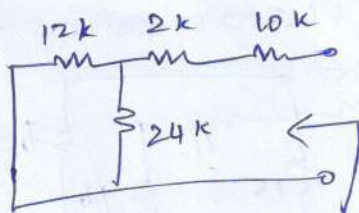


$$V_{th} = 12 + 8 + 20 = 40V$$

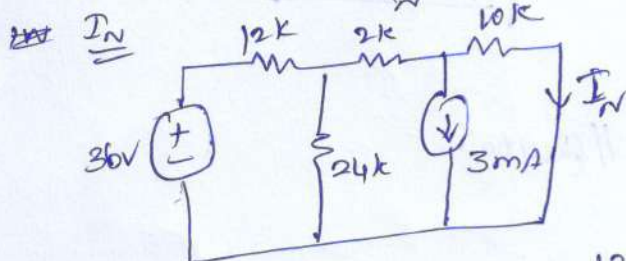
For max power to be transfered $R_L = R_{th} = 12 \Omega$.

$$\text{Power delivered} = \frac{V_{th}^2}{4R_{th}} = \frac{(40)^2}{4 \times 12} = 33.33 W$$

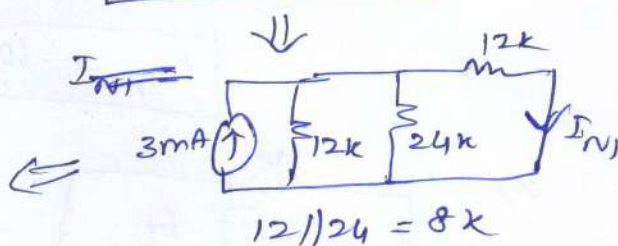
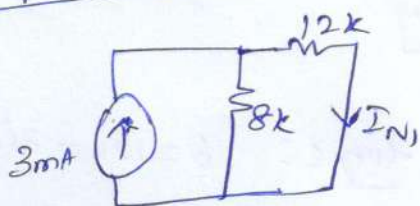
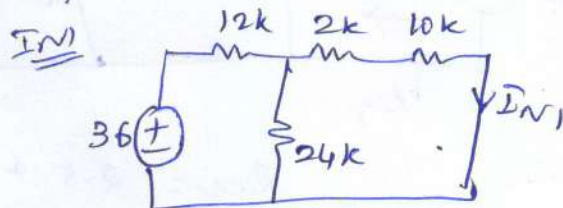
4) Norton's eq.



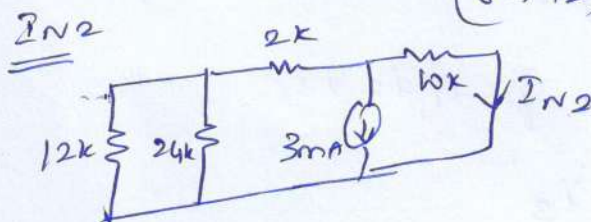
$$R_N = 12k + (2k \parallel 10k) = 20k.$$



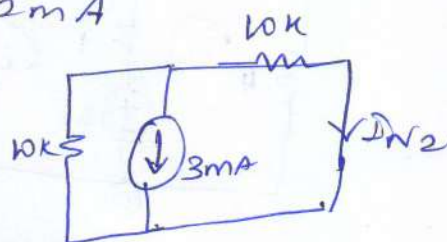
using SPT $I_N = I_{N1} + I_{N2}$



$$I_{N1} = \frac{3 \times 10^{-3} \times 8 \times 10^3}{(8 + 12) \times 10^3} = 1.2 \text{ mA}$$



$\Rightarrow$



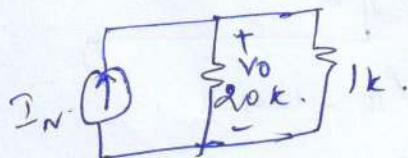
$$2k + 12k \parallel 24k = 10k.$$

$$\therefore I_{N2} = -1.5 \text{ mA}.$$

$$\therefore I_N = (1.2 - 1.5) \text{ mA}$$

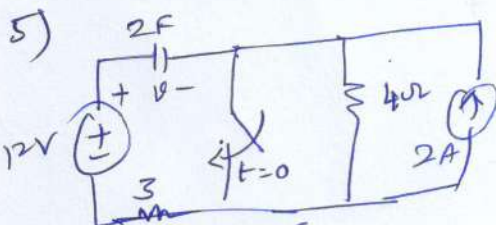
$$I_N = -0.3 \text{ mA}.$$

Norton's eq.



$$\therefore V_0 = (1k) \left(\frac{20}{20+1} \right) (-0.3 \times 10^{-3})$$

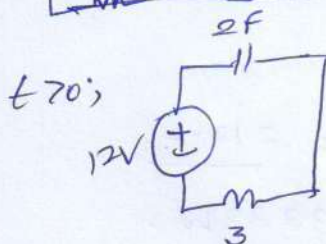
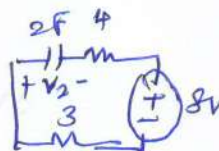
$$= -285.7 \text{ mV}$$



$t < 0$; $V = V_1 + V_2$ (SPT)

Due to 12V source: $V_1 = 12V$

Due to 2A source $\rightarrow$ do source transformation
 $V_2 = -8V.$



$$V(t) = V(\infty) + [V(0) - V(\infty)] e^{-t/\tau}$$

$$V(t) = V(\infty) + [V(0) - V(\infty)] e^{-t/\tau}$$

$$V(\infty) = 12 \quad V(0) = 4. \quad \tau = RC = 6$$

$$V(t) = 12 + (4 - 12) e^{-t/6}$$

$$V(t) = 12 - 8 e^{-t/6} \text{ V}$$